

CSE 4408
System Analysis and Design Lab

Final Project Report

Project Title: Female Hall Management System

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Executive Summary

This project report presents the comprehensive system analysis and design of the Female Hall Management System developed for the Islamic University of Technology (IUT). The project addresses the existing inefficiencies and challenges in hall administration, which currently relies on manual operations for student registration, room assignment, leave approvals, and complaint management. The traditional system results in time delays, data redundancy, loss of information, and communication gaps between residents and hall authorities. To overcome these issues, the proposed FHMS offers a digital, integrated, and user-friendly solution that modernizes the overall hall management process.

The Female Hall Management System is designed as a web-based platform that automates key hall operations. It enables digital record-keeping, automated room allocation, online leave and complaint submission, and efficient fee tracking. The system ensures transparency, accessibility, and efficiency by allowing both hall staffs and residents to interact with the platform in real time. Students can access personal information, submit requests, and view status updates, while administrators can monitor operations through dashboards, generate reports, and handle exceptions effectively.

A detailed Technical, Economic, and Operational (TEO) feasibility analysis was conducted to ensure the project's practicality within the university's infrastructure. The technical feasibility confirmed that the system could be developed using open-source web technologies like HTML, CSS, PHP, and MySQL, which are compatible with IUT's existing network environment. The economic feasibility demonstrated cost-effectiveness, as the development relies primarily on student labor and free software tools, minimizing financial investment. The operational feasibility indicated a high level of adaptability, as both hall staff and students are accustomed to using digital systems.

Following an Agile development methodology, the project was structured into iterative sprints to incorporate feedback from users throughout development. Prototyping and user stories were used to refine the system's functionality, ensuring alignment with stakeholder expectations. Key system models—including Data Flow Diagrams (DFDs), Data Dictionary, Process Specifications, and Output Designs—were developed to maintain design clarity and consistency.

The final FHMS design introduces real-time dashboards for administrators, digital forms for residents, and automated notifications for important updates such as leave approvals and fee reminders. By integrating automation and usability principles, the system significantly enhances data accuracy, reduces workload, and fosters a smart, paperless, and student-centric hall management environment at IUT.

1. Introduction and Project Initiation

a. Organizational Context: The Islamic University of Technology (IUT) operates several student halls to accommodate female and male residents. The Female Hall serves as both a residential and administrative facility for female students. The Hall Office manages student data, room allocations, complaint resolutions, and leave permissions manually, resulting in delays and inconsistencies. The *Female Hall Management System (FHMS)* aims to digitalize these processes, making management more efficient and transparent.

b. Problem Statement & Justification:

Problem:

The existing hall management relies heavily on paperwork and verbal communication, which often causes delays, data loss, and confusion among residents and staff.

Objectives:

1. Automate room and leave management.
2. Improve complaint tracking and resolution.
3. Enhance transparency and accountability through digital records.
4. Provide real-time updates for both residents and staffs.

Justification:

The project aligns with IUT's digital transformation goals and offers a sustainable, long-term solution with minimal resource investment.

c. Feasibility Analysis:

1. Technical Feasibility:

The system can be developed using standard web technologies (HTML, CSS, JavaScript, PHP/MySQL). The IUT IT infrastructure supports web hosting.

2. Economic Feasibility:

The project requires minimal cost—student labor and open-source software. The long-term benefits include time savings and reduced manual workload.

3. Operational Feasibility:

Staffs and students are familiar with digital interfaces, ensuring smooth adoption. Minimal training will be needed.

Break-even & Payback:

The project is expected to break even within 1 semester due to time and resource

savings.

Component	One-Time (BDT)	Recurring(BDT/year)
Web Portal Dev	6,00,000	Null
User Training	2,00,000	Null
Security & Backup	Null	1,00,000
System Maintenance	Null	2,00,000
Hardware Setup	3,00,000	Null

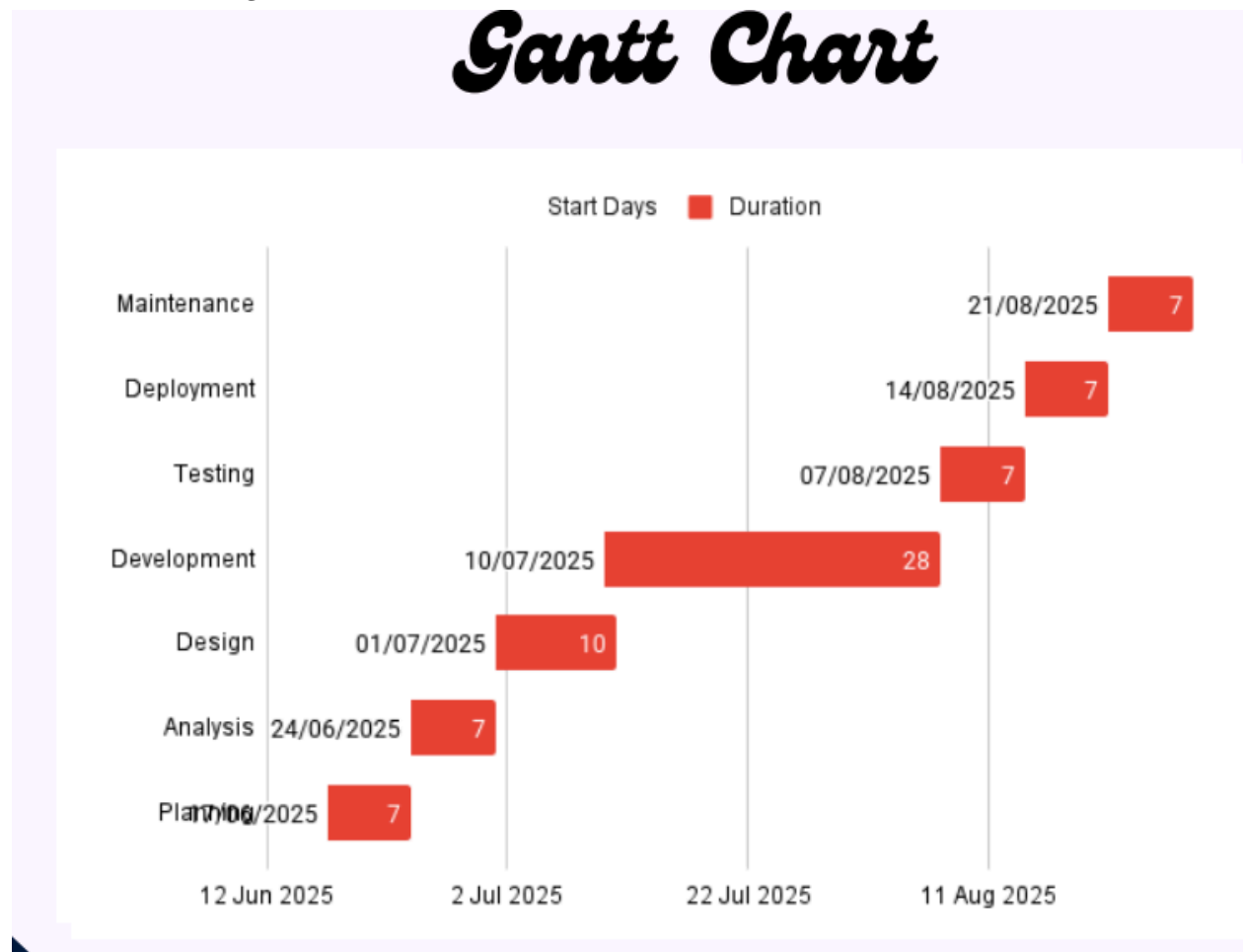
d. **Project Management Plan:**

WBS Phases: Requirement analysis → Feasibility → Design → Development → Testing → Documentation → Deployment.

Final version of WBS

<i>Work Breakdown Structure & Timeline</i>			
Phase	Sub-Activities	Time	Justification
Planning	Scope, Stakeholders	1 wk	Small team, clear scope
Analysis	User stories	1 wk	Agile quick story creation
Design	UI mockups, DB design	1.5 wk	Moderate UI & DB work
Development	Token system, User login	2 wk	Core features development
	Leave & Complaint forms	2 wk	Additional modules
Testing	Unit, Integration, UAT	1 wk	Tests during sprints + UAT
Deployment	Hosting, Training	1 wk	Quick deployment & onboarding
Maintenance	Bugfix, Feedback	1 wk	Post-launch support

Our timeline using Gantt chart from Lab 4.



Risk Analysis:

Technical risk: Limited backend expertise → mitigated through online tutorials.

Time risk: Task overlap → managed via weekly scrum meetings.

Data loss: Regular backups during development

2. Requirements Investigation and Analysis

a. Information Gathering Strategy:

We used both **interactive** and **unobtrusive** methods:

- **Interviews** with hall staff and residents to identify workflow issues.
- **Questionnaires** distributed among 15 students to collect preferences (e.g., mobile vs web interface).
- **Observations** of the current manual process to note delays.
- **Sampling** of leave applications and complaints for error trend analysis.

These methods provided both qualitative and quantitative insights for accurate requirement formulation.

b. Summary of Findings:

Key Issue	Stakeholder's feedback	Proposed Feature
Slow leave approval	Manual tracking delays	Online leave request and approval
Poor complaint follow-up	No structured record	Complaint submission and tracking
Room confusion	Paper-based records	Automated room allocation
Fee tracking issues	Manual calculation	Automated fee computation
Lack of updates	No alert system	Email/SMS notifications

c. Consolidated System Requirements:

Functional Requirements:

- 1.The system shall allow residents to register and log in.
- 2.The system shall enable staff to assign rooms.
- 3.The system shall allow students to submit leave requests and complaints.
- 4.The system shall generate summary reports for administrators.

Non-Functional Requirements:

- 1.System availability: 99%.
- 2.Response time: <3 seconds.
- 3.Secure authentication via student ID.
- 4.Mobile-responsive interface.

3. System Modeling and Design

Agile Modeling & Prototyping:

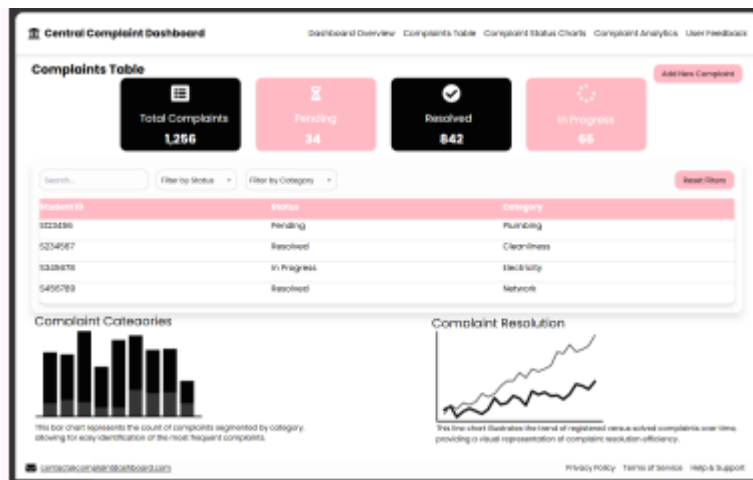
User Stories:

A. *As a student*, I want to fill online leave approval form so that I don't have to fill the form manually.

B. As a clerk, I want to assign rooms efficiently to avoid overlap.

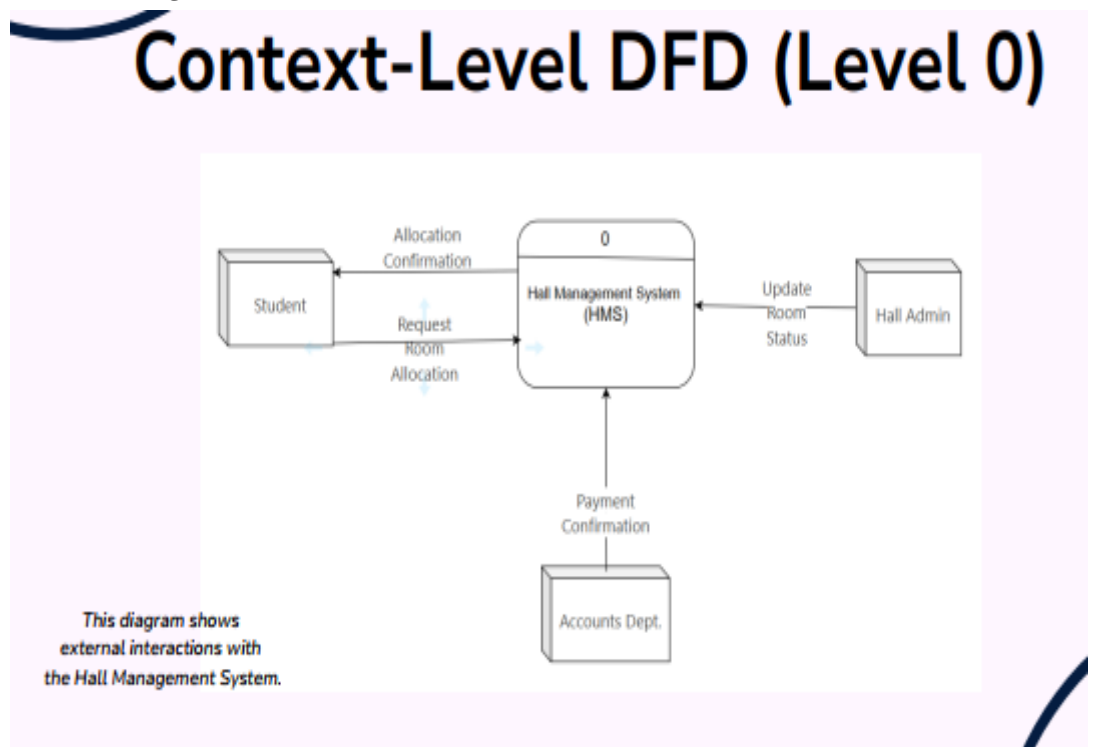
C. As a warden, I want to track complaints in real-time for faster response.

Prototyping in Figma showed a simple dashboard for each user role (Student, Clerk, Warden). Central complain dashboard prototype

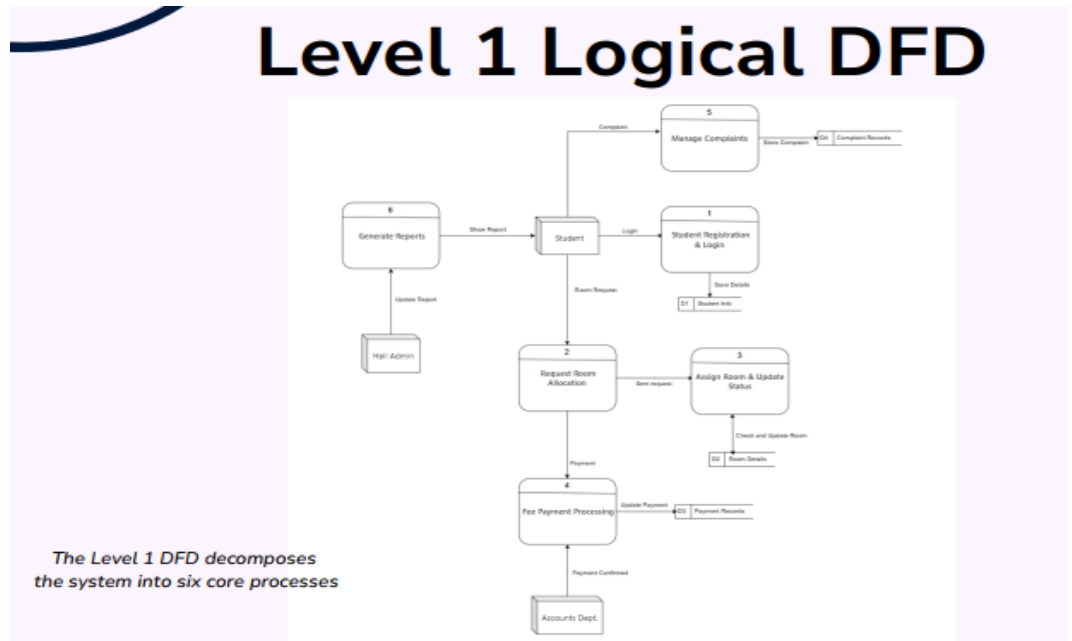


a. Process Modeling (DFDs): Includes:

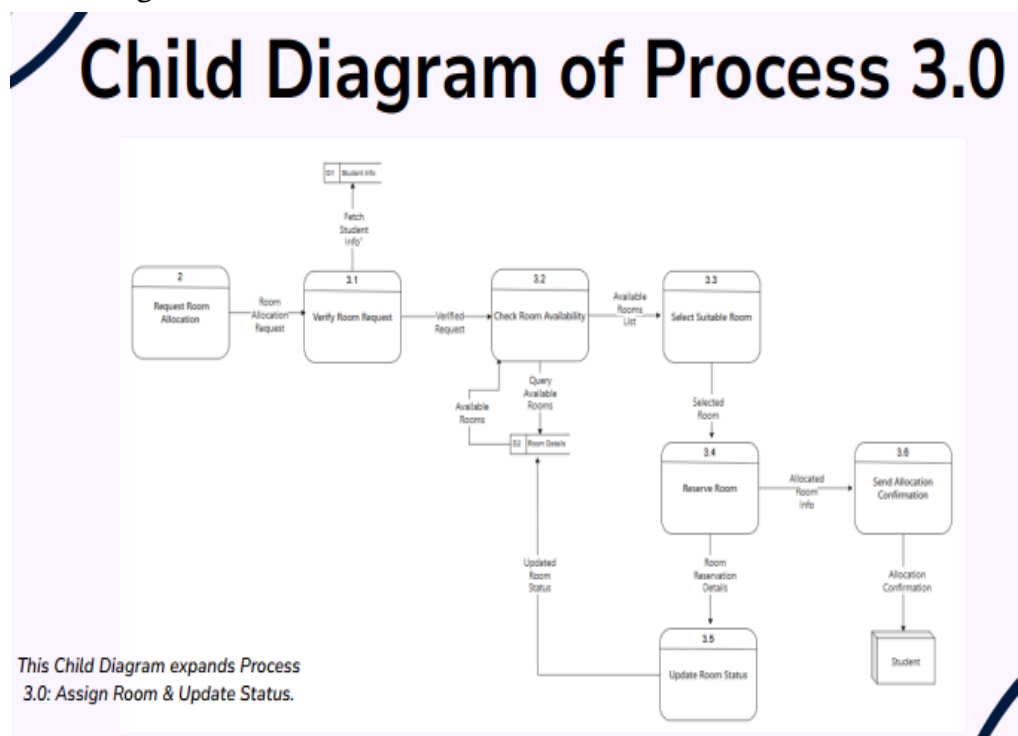
- Context Diagram (FHMS overview)



- Level 0: Major processes — Leave Management, Complaint Handling, Fee Calculation.



- Child Diagram:



b. Data Modeling (Data Dictionary):

Defined all data flows and structures:

- Data Flows: Leave Request, Complaint Data, Room Info, Fee Record.
- Data Stores: Student Record, Room Database, Complaint Log.
- Elements: StudentID, RoomNo, LeaveStatus, ComplaintType, FeeAmount.

c. Process Logic Specification:

1. Assign Room to Student (DFD 1.1)

Technique: Structured English

IF AvailableRooms for Requested Room Type is empty THEN REJECT "Requested room type not available" ADD StudentID to waiting list NOTIFY student about unavailability ELSE ASSIGN first available room to StudentID UPDATE RoomStatus to "Occupied" GENERATE RoomAssignmentConfirmation for StudentID END

2. Hall Fee Payment (DFD 2.1)

Technique: Decision Table

Rule	Amount Paid Correct?	On Time?	Scholarship?	On Hall List?	Actions
R1	-	-	Yes	Yes	Payment Not Required
R2	Full	Yes	No	Yes	Accept Payment
R3	Full	Late	No	Yes	Reject Payment + Notify Student
R4	Incorrect	-	No	Yes	Reject Payment + Notify Student
R5	Paid (shouldn't)	-	Yes	Yes	Refund Payment + Notify Student

3. Route Complaint

Technique: Decision Tree



d. Output Design:

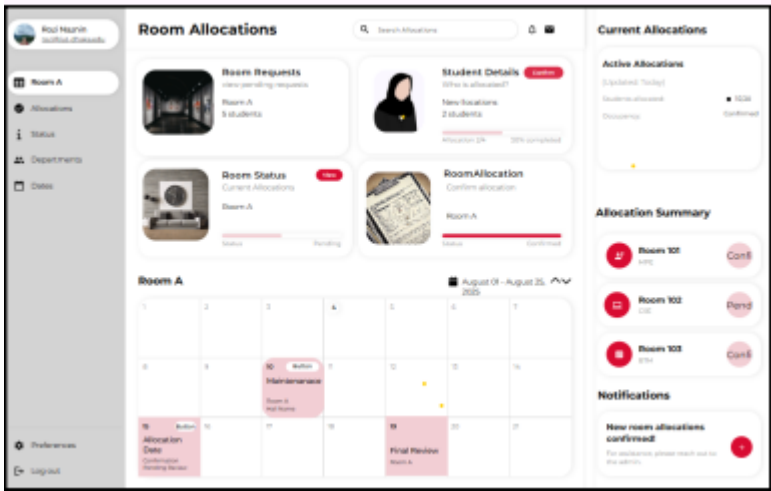
1. Room Allocation Dashboard

User: Warden / Admin

Purpose: View and manage room occupancy.

Design: Simple table with search and filter options; color-coded rooms (vacant/occupied) for quick status check.

Justification: Enhances visibility, minimizes search time, supports quick decision.



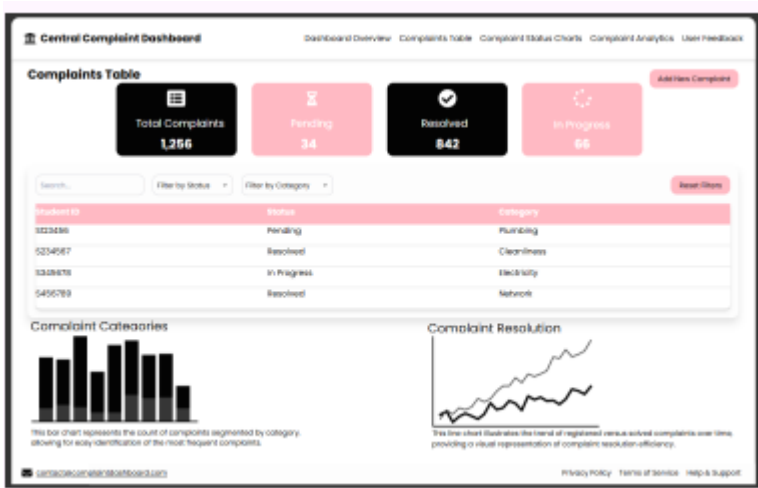
2. Central Complaint Dashboard

User: Warden / Staff

Purpose: Monitor and respond to complaints.

Design: Summary counters, tabular list, bar and line charts by category and status.

Justification: Visual overview aids quick tracking and efficient issue resolution.



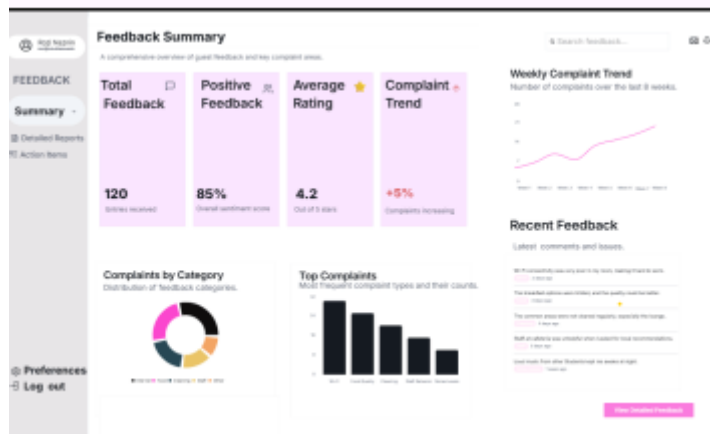
3. Feedback Overview Dashboard

User: Warden / Admin

Purpose: Analyze complaint and feedback trends.

Design: Charts highlighting top issues with flat icons and simple color scheme.

Justification: Easy to interpret trends; supports proactive management decisions.



4. Self-Assessment

- a. **Complex Engineering Problem (CEP) Analysis:** The project tackled a multifaceted problem that involved balancing several conflicting requirements such as accuracy versus usability, and automation versus manual control. Designing the *Female Hall Management System* required analyzing real-world workflows, understanding stakeholder expectations, and resolving operational inefficiencies within the hall environment. The problem also demanded integration of technical, social, and organizational perspectives—ensuring that automation improved efficiency without compromising human oversight or data privacy. Through iterative requirement gathering, feasibility assessment, and modeling, the team effectively addressed the complexity of developing a robust and user-centered information system.
- b. **Complex Engineering Activity (CEA) Demonstration:** The project involved executing complex engineering activities requiring careful planning, coordination, and decision-making. Our team managed diverse technical and human resources, distributed tasks using Agile sprints, and continuously refined goals based on stakeholder feedback. Considerations of ethical responsibility—particularly regarding the handling of sensitive information related to female students—were incorporated into both design and implementation. Collaboration, sprint retrospectives, and structured documentation ensured traceability, quality assurance, and consistency across development stages, demonstrating an integrated engineering approach.
- c. **Knowledge Profile (WK) Application:** Throughout the project, the team applied a wide range of engineering knowledge and skills. Software engineering fundamentals (WK3) were employed for requirement analysis, modeling, and system design. Specialist knowledge (WK4) in database management and structured modeling guided the creation of the data dictionary and process logic specifications. Resource design and management principles (WK5) were applied in

project scheduling, risk analysis, and task allocation. Lastly, engineering practice (WK6) was demonstrated through the iterative use of Agile methodologies, prototype evaluations, and stakeholder communication—resulting in a practical and sustainable software solution.

5. Conclusion and Reflection

This final section summarizes our work and reflects on the learning process.

- a. **Project Summary:** The Female Hall Management System (FHMS) successfully digitalizes and automates key hall management operations such as room assignment, complaint resolution, leave requests, and fee processing. By replacing manual paperwork with a centralized, web-based platform, the system enhances operational efficiency, transparency, and accessibility for both hall authorities and students. The solution improves communication, reduces administrative delays, and ensures data accuracy, contributing to a smoother residential experience for female students at IUT. Overall, FHMS provides a sustainable, secure, and user-friendly approach to hall administration, aligning with the university's vision for digital transformation.
- b. **Team Reflection:** The development of this project was carried out using Agile methodology, allowing flexibility and continuous improvement through iterative design and user feedback. Weekly meetings and sprint reviews ensured effective communication and task coordination among team members. The team faced challenges in time management, workload balancing, and technical troubleshooting, especially while integrating multiple features and maintaining consistency across design artifacts. However, strong collaboration, clear role division, and the use of prototypes helped overcome these challenges. This project enhanced our understanding of real-world system development, teamwork dynamics, and the application of system analysis principles in practice.
- c. **Future Work:** While the current version of FHMS fulfills core functional requirements, several enhancements can be incorporated in future iterations to further increase its effectiveness and usability:
 1. **Integrate biometric attendance tracking** to ensure secure and automated monitoring of hall residents.
 2. **Implement mobile notifications via SMS or app alerts** for real-time communication regarding approvals, fee reminders, and updates.
 3. **Introduce data analytics dashboards** to visualize occupancy rates, complaint trends, and operational efficiency over time.

These future developments would expand the system's capability, improve user engagement, and support data-driven decision-making for hall administration.

6. Appendices (Optional)

Interview questionnaires:

- a. What kind of updates would you like to receive from the hall (e.g., notices, request
- b. status)?

- c. Would you prefer receiving notifications by SMS, email, or dashboard alert?
- d. How comfortable are you using a web-based form to submit requests?
- e. What features would make the system easier for both students and hall staff?
- f. Can you describe a recent case where your request was delayed or lost?